

GradeWise

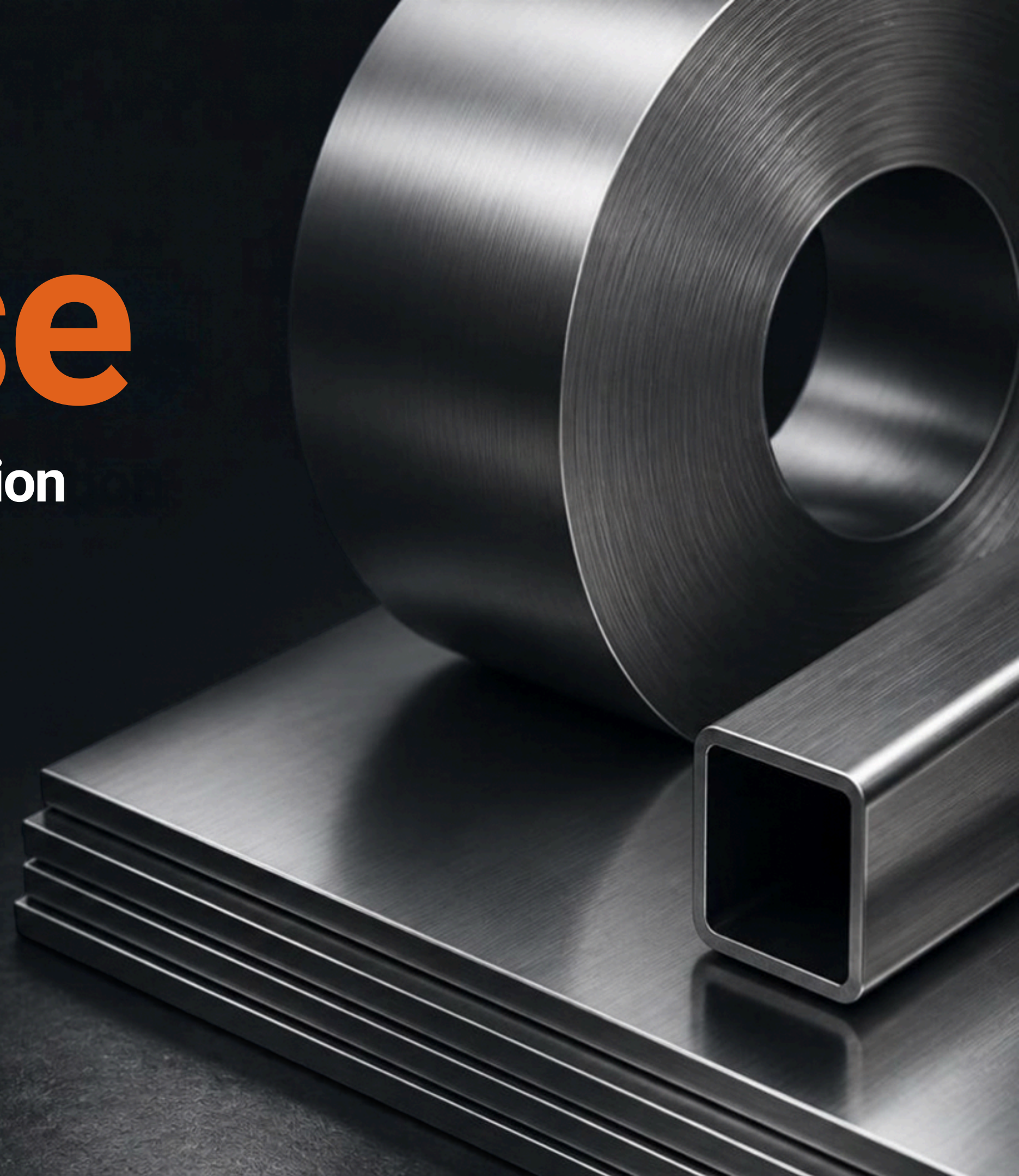
AI-Guided Stainless Steel Grade Selection

github.com/Crayon-100/Gradewise

Stainless Spark 2026 • Problem Statement 2

Jindal Stainless

Every grade, **argued**.
Every number, **proven**.



THE PROBLEM

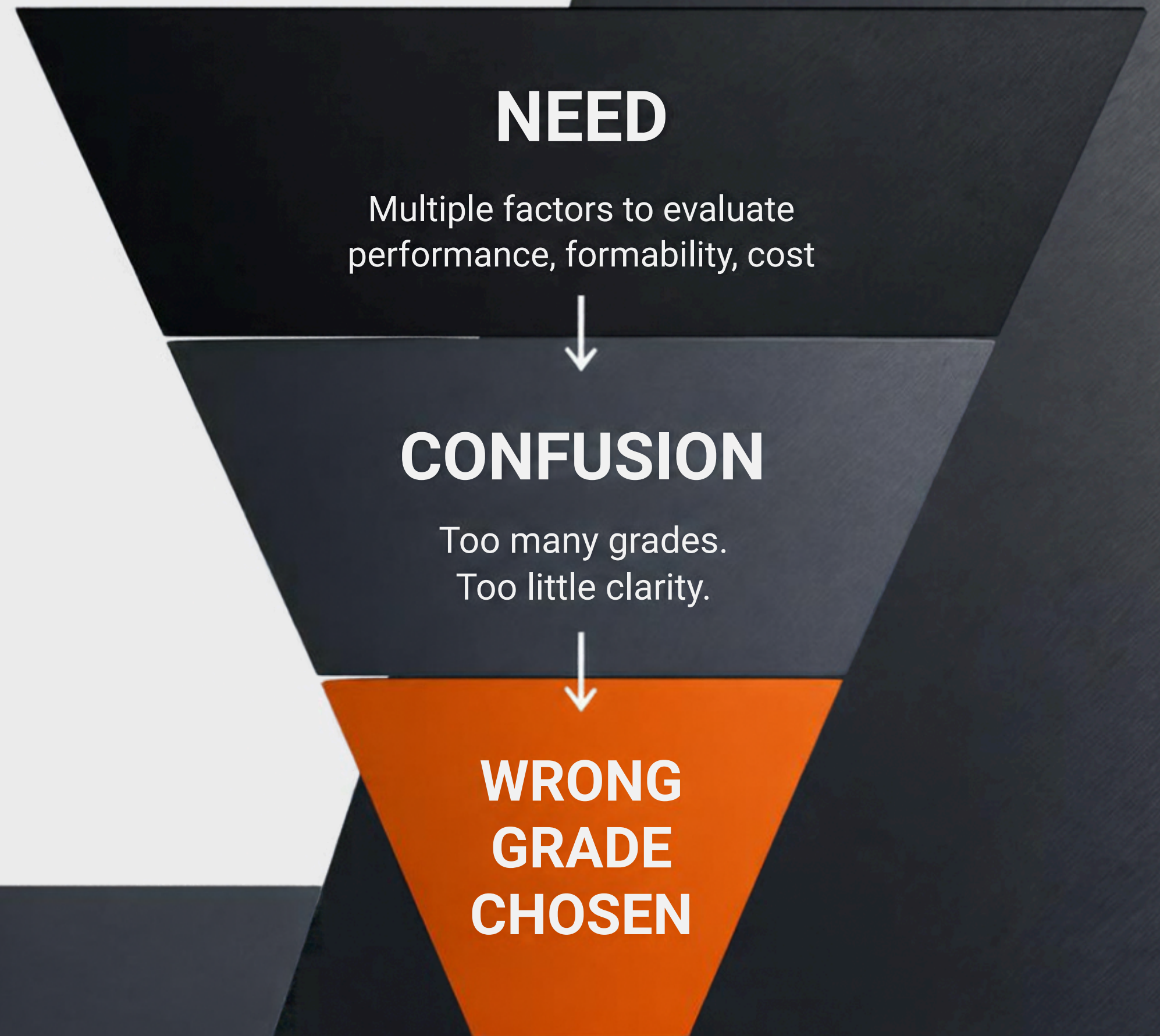
**Grade selection is complex.
The wrong choice is costly.**

Engineers, fabricators and first-time buyers
balance performance, formability and cost

Lack of guidance leads to over-specified or under-
specified

Every decision impacts cost efficiency
and long-term reliability

**The outcome: higher costs, lower
reliability, lost competitiveness.**



THE SOLUTION

AI reasoning. Physics precision. Visual clarity.
Smarter, faster, more confident grade selection.



Deterministic requirement-encoding algorithm



Server-side dataset constraint: AI can select and explain only grades present in the validated dataset



Same normalized requirements → same recommendations; semantically equivalent prompts are normalized before selection.



AI selects grades and writes explanations; the deterministic physics module generates every numerical result.

01 Describe Your Need

I need a corrosion resistant sheet for a food processing tank, 2 mm thick, used in coastal environments.

Plain language input.
No jargon required.

02 AI Shortlists Best Grades

Recommended Grades

316L

Best overall corrosion resistance; higher cost

304L

Good corrosion resistance; lower cost

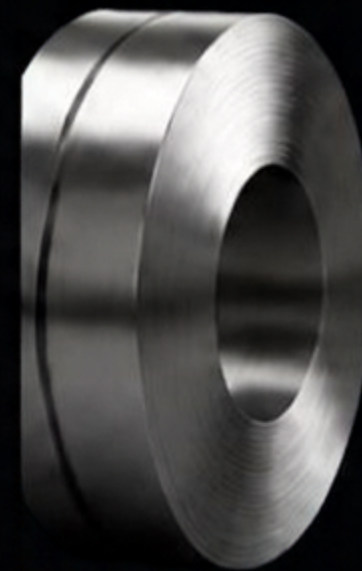
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High strength; better pitting resistance

- Structured requirement encoding converts each plain-language need into a canonical set of constraints, so equivalent inputs resolve to the same grade recommendations.

- Recommendations are constrained to the server-side stainless grade dataset, not free-form model output.

03 Physics Engine Calculates



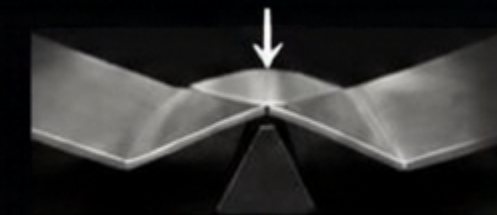
Load
2.45 kN

Deflection
1.28 mm

Tensile Strength
620 MPa

Deterministic physics engine delivers exact performance metrics.

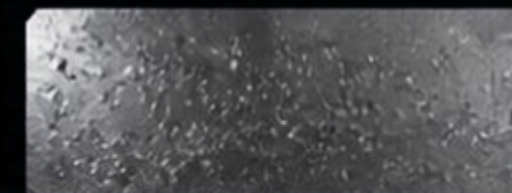
04 Interactive Visuals



See-saw bend test



Hanging-weight tensile test



Corrosion-resistance visual

Interactive tests bring material behavior to life with visual understanding.

TECHNICAL DEPTH & ARCHITECTURE

A hybrid system combining deterministic engineering and AI reasoning

MODERN FRONTEND

Next.js 15 App Router
TypeScript, Tailwind CSS
Frame Motion

ROBUST BACKEND

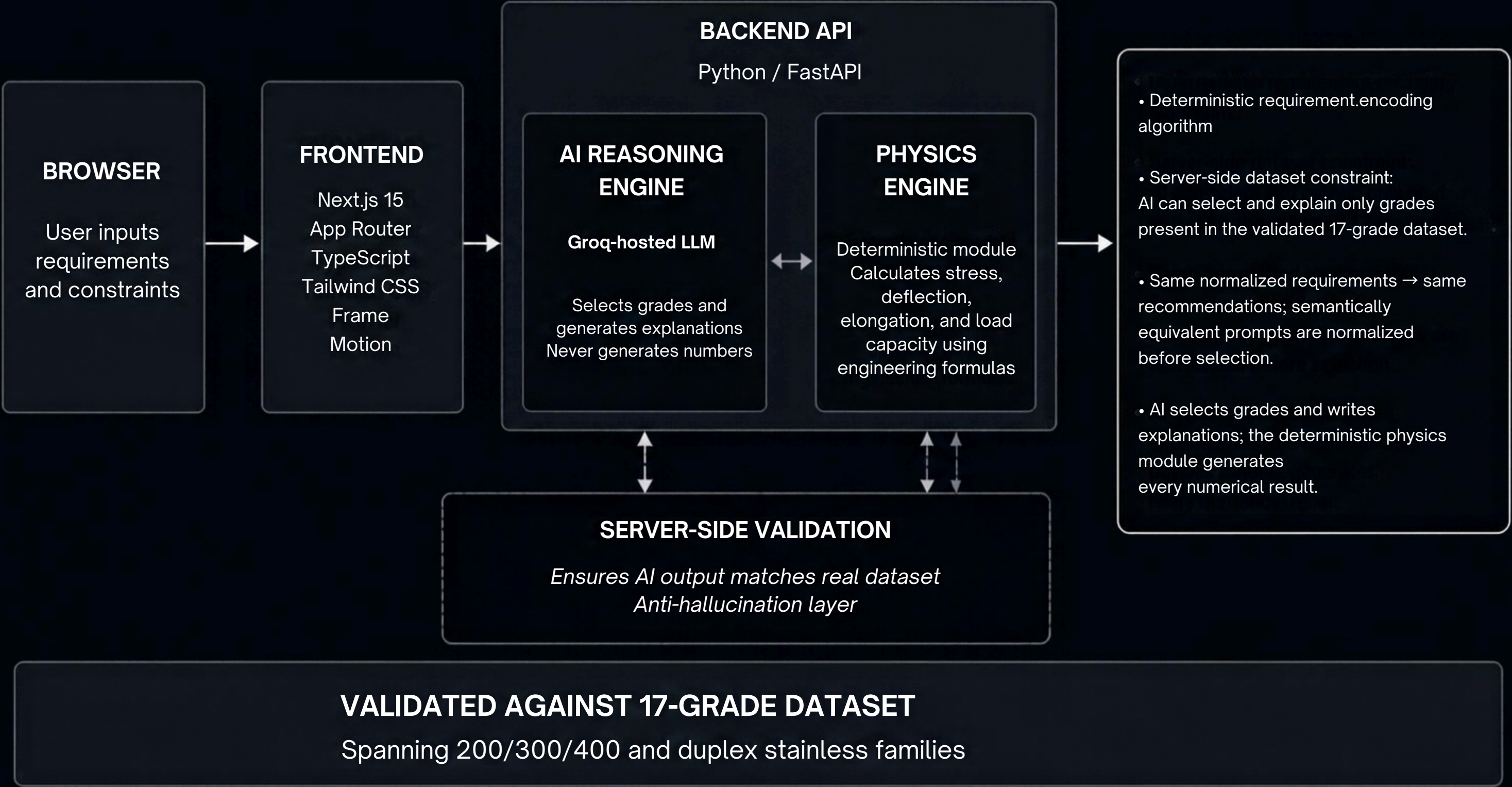
Python / FastAPI
RESTFUL API

AI WITH GUARDRAILS

Groq-hosted LLM selects grades and writes explanations.
Never generates numbers.

DETERMINISTIC PHYSICS

Computes stress, deflection, elongation, load capacity using standard engineering formulas.



INNOVATION

What Makes This Different

	TRADITIONAL SELECTION	GRADEWISE
SPEED	Manual lookup in static tables and PDFs	Conversational trade-off reasoning delivers instant, context-aware answers
TRUST IN NUMBERS	Black-box outputs with limited traceability	AI judgment is separated from deterministic calculations for verifiable results
ACCESSIBILITY	Technical formats require specialist expertise	Interactive bend and tensile visualizations make materials science intuitive
TRADE-OFF TRANSPARENCY	Trade-offs buried across documents and notes	Clear reasoning of compromises across performance, formability, and cost

FEASIBILITY

**Built today. Validated engineering.
Extensible by design.**

DEPLOYED NOW

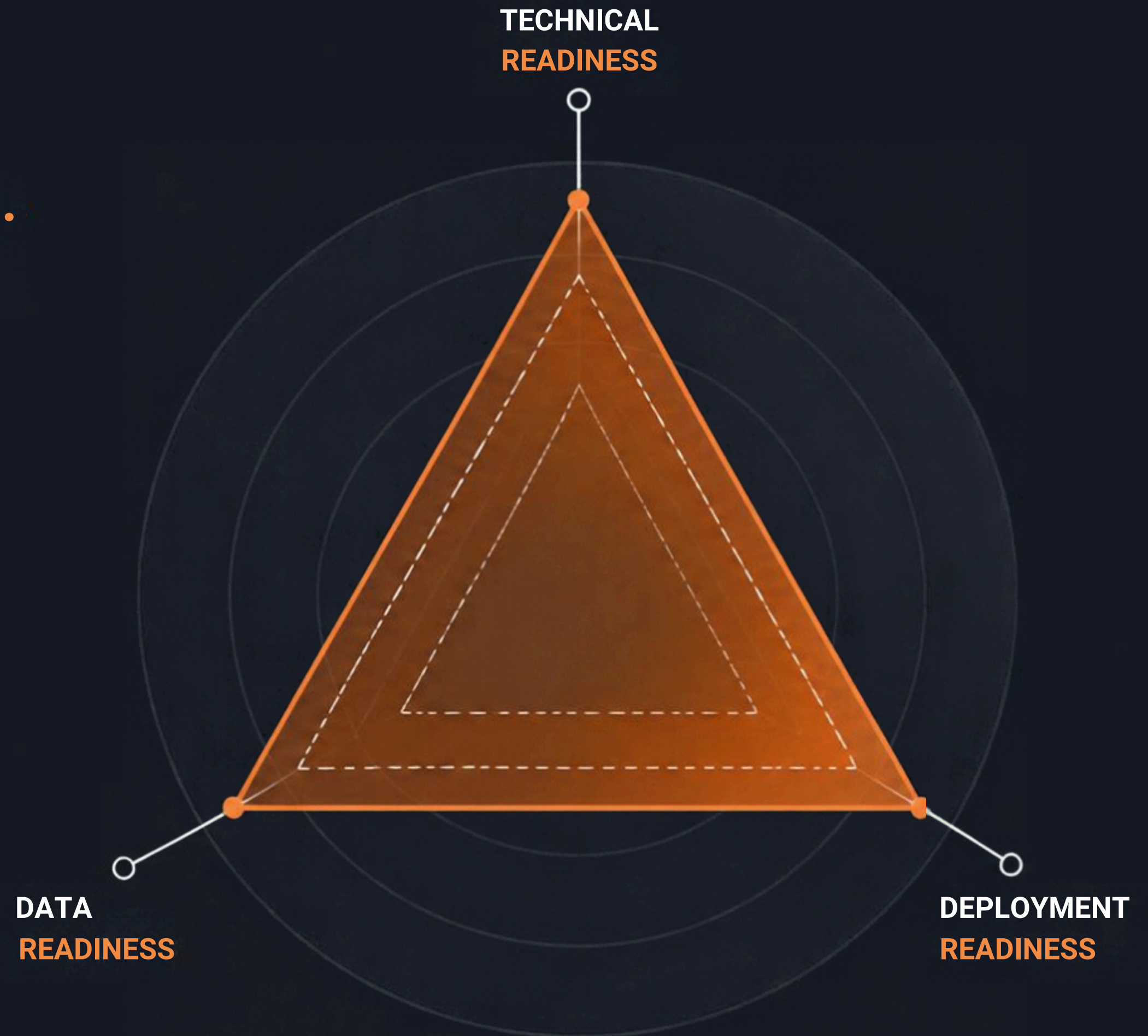
Live on free Vercel frontend and
Render backend, near-zero infra cost

ENGINEERING VALIDITY

Uses standard published engineering equations
cross-checked against real load-capacity references

DATA EXTENSIBILITY

Adding a new grade =
adding one row, no re-architecture



SCALABILITY ROADMAP

Expanding capability, data, and impact in three phases.

01

NOW

Expand beyond rods and bars
to sheets, plates, and tubes

Add corrosion and
hardness simulations

02

NEXT

Replace demo dataset
with Jindal's full certified
grade catalog

Integrate mill-test-certificate
data for complete accuracy

03

LATER

Embed within Jindal sales
and technical workflows

Offer as a white-label tool
for other manufacturers

IMPACT

FOR CUSTOMERS

Faster, confident material decisions
Fewer over-specified
Fewer in-service failures

FOR JINDAL

A differentiated digital touchpoint
reducing sales-engineering workload

FOR SUSTAINABILITY

Correctly matched grades reduce
unnecessary material use and costs

Every grade, argued. Every number, proven.

